



EDUCATION

Ph. D. in Cognitive Science with a focus on Deep Neural Representation Learning Universität Osnabrück Research Training Group in Computational Cognition (DFG-funded)	11/2025 - 11/2021
M. Sc. in Math. Sciences African Institute for Mathematical Sciences AIMS / Mastercard Foundation Scholarship	09/2020 - 07/2021
M. Sc. in Computer Science & Engineering University of Dhaka Queen Elizabeth Commonwealth Scholarship	07/2018 - 08/2020
B. Sc. In Computer Science University of Makeni Undergraduate Merit Scholarship	09/2012 - 02/2017

FURTHER EDUCATION

AI for Medicine Specialization DeepLearning.AI	2026
AI for Healthcare Specialization Stanford	2025
Deep Learning Bootcamp MUST, North-West University, SA	2025
Confidentiality and Data Protection in Health Research CITI	2025
AI Engineering Prof. Certificate IBM	2023
Data Science Prof. Certificate IBM	2023
Python Programming and Data Structures Coursera, University of Michigan	2023
Mediterranean Machine Learning School Università Milano Bicocca, Italien	2022
Deep Learning Program Neuromatch Academy	2022

I am a **Biomedical AI Scientist** with experience in medical image analysis, multi-modal clinical AI, explainable AI, and robust representation learning. My work focuses on developing clinically meaningful machine learning systems for image data and structured health data, with a particular emphasis on generalization, interpretability, and clinical applicability.

My **research interests** include: medical image analysis; multimodal clinical AI; trustworthy and explainable AI; robustness and generalization; representation learning; clinical decision support; resource-efficient AI in healthcare.

PROFESSIONAL EXPERIENCE

Co-founder and Biomedical AI Researcher AI in NTD Consortium / Osnabrück	12/2024 – present
- Co-founded and lead a research consortium with 33 interdisciplinary members from 21 institutions - With the goal of establishing a global interdisciplinary research network leading to the creation of Hubs of Excellence - Coordination of work packages and milestones for members - Roadmap creation and management - Coordination and revision of draft manuscripts	
Research Assistant (PhD) University of Osnabrück / Osnabrück	11/2021 – 12/2024
- Research in deep representation learning for AI models, including self-supervised learning methods such as contrastive, generative, and joint-embedding learning - Development of joint-embedding, contrastive, and generative models for disentangled representation learning, such as Barlow Twins, VAEs, or supervised contrastive learning - Improvement of data processing efficiency by at least 20% and model accuracy by at least 25% for model target tasks - Curation of the SynSpeech datasets and maintenance of open-source access - Presentations and seminars to showcase your own research, e.g., NeurIPS (2023), German Conference on Artificial Intelligence Doctoral Consortium (2023), ICLR (2024) - Supervision of master's students' research	
Research Assistant University of Makeni / Makeni, Sierra Leone	02/2017 – 07/2018
- Taught 4 different computer science courses with approximately 40 students per course - Developed course materials, homework assignments, and exams	
Lecturer Limkokwing University of Creative Technology / Freetown, Sierra Leone	01/2018 – 07/2018
Taught courses on programming logic and design for approximately 120 students in classes of approximately 40 students	

HARD SKILLS

Machine Learning / Deep Learning

PyTorch, TensorFlow/Keras, scikit-learn, Transformers

Data Manipulation & Visualization Libraries

NumPy, Pandas, Matplotlib, Seaborn

Programming Languages

Python, R, C/C++, SQL

Systems

Linux; Windows, High Performance Computing with SLURM (HPC), Grid Computing with Oracle Sun Grid Engine, Docker, GCP, GPU Computing

Neuroimaging Data & Tools

BIDS, FSL, FreeSurfer

Medical Image Processing

OpenCV, SimpleITK, NiBabel, Pydicom, ANTsPy, scikit-image

Explainable, Fair, and Trustworthy AI

SHAP, Captum, Fairlearn, AIF360, LIME, TorchCAM, iNNvestigate

SOFT SKILLS

Organization

Project Management

Communication

Personal Responsibility

Work Independently

Solution-Oriented Problem Solving

Flexibility and Adaptability

Teamwork and Collaboration

LANGUAGES

English	Native-level proficiency
German	Intermediate
French	Basic
Krio	Native
Kono	Native

PUBLICATIONS (SELECTED)

Lead Author

- Brima Y. *The Case for Multimodal Causal Representation Learning in Clinical Artificial Intelligence*. Under review at *Nature Machine Intelligence*
- Brima Y, Atemkeng M. *Explainable Multimodal Hybrid CNN-Transformer Models for Alzheimer's Disease Diagnosis*. Submitted to *npj Digital Medicine*.
- Brima Y, Atemkeng M, Kallon LH, Niyukuri D, Vacavant A, Saidu S, Chen D-G. *Few-shot Learning with Cross-country Generalization of Tabular Machine Learning and Foundation Models for Childhood Anemia Prediction under Distribution Shift*. Under review at *Nature Machine Intelligence*.
- Brima Y, Atemkeng M, Ngueajio M, et al. *Artificial Intelligence in Neglected Tropical Diseases: Current Applications, Challenges, and Opportunities*. Under review at *npj Digital Medicine*.
- Brima, Y., Atemkeng, M. An empirical study of machine learning robustness and scalability for imbalanced tabular clinical data in emergency and critical care. *Sci Rep* 16, 18004 (2026).
- Brima Y, Krumnack U, Pika S, Heidemann G. *Understanding Self-Supervised Learning of Speech Representation via Invariance and Redundancy Reduction*. Information. 2024 Feb 15;15(2):114.
- Brima Y, Krumnack U, Pika S, Heidemann G. *Learning Disentangled Audio Representations through Controlled Synthesis*. Presentation at ICLR Tiny Papers Track 2024. arXiv:2402.10547.
- Brima Y, Atemkeng M. *Saliency-driven Explainable Deep Learning in Medical Imaging: Bridging Visual Explainability and Statistical Quantitative Analysis*. *BioData Mining*, 2024, 17(1):18.
- Brima Y, Krumnack U, Pika S, Heidemann G. *Learning Disentangled Speech Representations*. *New in Machine Learning Workshop, NeurIPS 2023*, New Orleans, USA.
- Brima Y, Atemkeng M, Tankio Djiokap S, Ebiele J, Tchakounté F. *Transfer Learning for the Detection and Diagnosis of Types of Pneumonia Including COVID-19 from Chest X-ray Images*. *Diagnostics*, 2021, 11(8):1480.
- Brima Y, Kamal Tushar MH, Kabir U, Islam T. *Deep Transfer Learning for Brain Magnetic Resonance Image Multi-class Classification*. *Dhaka University Journal of Applied Science & Engineering*, 2022.

Co-authored

- Kamga L, Atemkeng M, Brima Y, Tchoupe Tchendji M. *Adaptive 2D sampling via prediction entropy for volumetric medical image classification: application to Alzheimer's disease diagnosis*. Under review at *IEEE Transactions on Medical Imaging*.
- Nguezet PV, Atemkeng M, Fute ET, Brima Y. *Bridging Visual Saliency and Large Language Models for Self-explainable Deep Learning in Medical Imaging*. Under review at *BioData Mining*.
- Nhlapho W, Atemkeng M, Brima Y, Ndogmo J-C. *Transformers for 3D Medical Image Analysis: A Systematic Review of Architectural Innovations, Performance, and Clinical Applications*. Accepted at *Artificial Intelligence Review*.
- Singh D., Brima Y, Levin F, Becker M, Hiller B, Hermann A, Villar-Munoz I, Beichert L, Bernhardt A, Buerger K, Butryn M. *An unsupervised XAI framework for dementia detection with context enrichment*. *Scientific Reports*, 2025, 15(1):39554.
- Hamlomo S, Atemkeng M, Brima Y, Nunhokee C, Baxter J. *A Systematic Review of Low-rank and Local Low-rank Matrix Approximation in Big Data Medical Imaging*. *Neural Computing and Applications*, 2025.
- Nhlapho W, Brima Y, Atemkeng M, Ndogmo J-C. *Bridging the gap: exploring interpretability in deep learning models for brain tumor detection and diagnosis from MRI images*. *Information*, 2024, 15(4):182.

REFERENCES

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Rhodes University, South Africa

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Vision & AI Groups
Institute for Cognitive Science
University of Osnabrück, Germany

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TALKS AND INVITED PRESENTATIONS (SELECTED)

Multimodal Representation for Robustness and Generalization in Biomedical Applications, Max Delbrück Center for Molecular Medicine, Germany, 2026

Artificial Intelligence and Its Role in Shaping Our Future, University of Makeni Computer Science Alumni Forum, Sierra Leone, 2025.

Multimodal Federated Learning for Robust Lung Cancer Prognosis, Fraunhofer SCAI, Germany, 2025.

Trustworthy Healthcare AI for Mental Health Risk Prediction, Neurobiology Research Unit, Copenhagen University Hospital, Denmark, 2024.

Assessing Explainability in Deep Learning for Medical Image Analysis, Fraunhofer SCAI, Germany, 2024.

Causal Representation Learning, Computer Vision Colloquium, Osnabrück University, Germany, 2024.

Learning Disentangled Audio Representations through Controlled Synthesis, Notable ICLR Tiny Papers, 2024.

ACADEMIC ENGAGEMENT & MEMBERSHIPS

Member of the Master's Program in Mathematical Sciences in Mathematical Epidemiology, **Scholarship Selection** 2026

Committee of the African Institute for Mathematical Sciences and the Africa Health Collaborative

Member of the Deep Learning Indaba Kambule **Doctoral Award Committee** 2025

Co-organizer of the Bridging Biological and Artificial Neural Networks **workshop** 2022
and the Deep Representation Learning Seminar at the University of Osnabrück 2023/24

Peer review of manuscripts for Scientific Reports; BMC Medical Informatics and Decision Making; Healthcare Technology Letters; npj journals; ICLR; ICML; MusIML Workshop; Deep Learning Indaba. Since 2023

Approved access to data from Demographic and Health Surveys (DHS) for research purposes in the field of reliable disease diagnosis. 2026

Access authorization and training for **controlled biomedical datasets and platforms**, including **UK Biobank/MRC, PhysioNet, LONI IDA, and the German FDPG framework**. 2025

Member of the MICCAI Society Since 2023

Member of Black in AI Since 2022

Member of Deep Learning Indaba Since 2022

Member of IEEE Since 2019

Member of AIMS Alumni Netzwerk Since 2021

SCHOLARSHIPS

CIFAR Deep Learning and Reinforcement Learning (DLRL) Summer School Inclusive AI Fully-funded Scholarship, Canada, 2024

Deep Learning Indaba Fully-funded Scholarship, 2023

Cambridge Ellis Machine Learning Summer School Fully-funded Scholarship, 2022

AIMS / Mastercard Foundation Scholarship 2020 – 2021

Queen Elizabeth Commonwealth Fully-funded Scholarship, 2018 - 2020

Undergraduate Merit Scholarship, 2013 – 2017